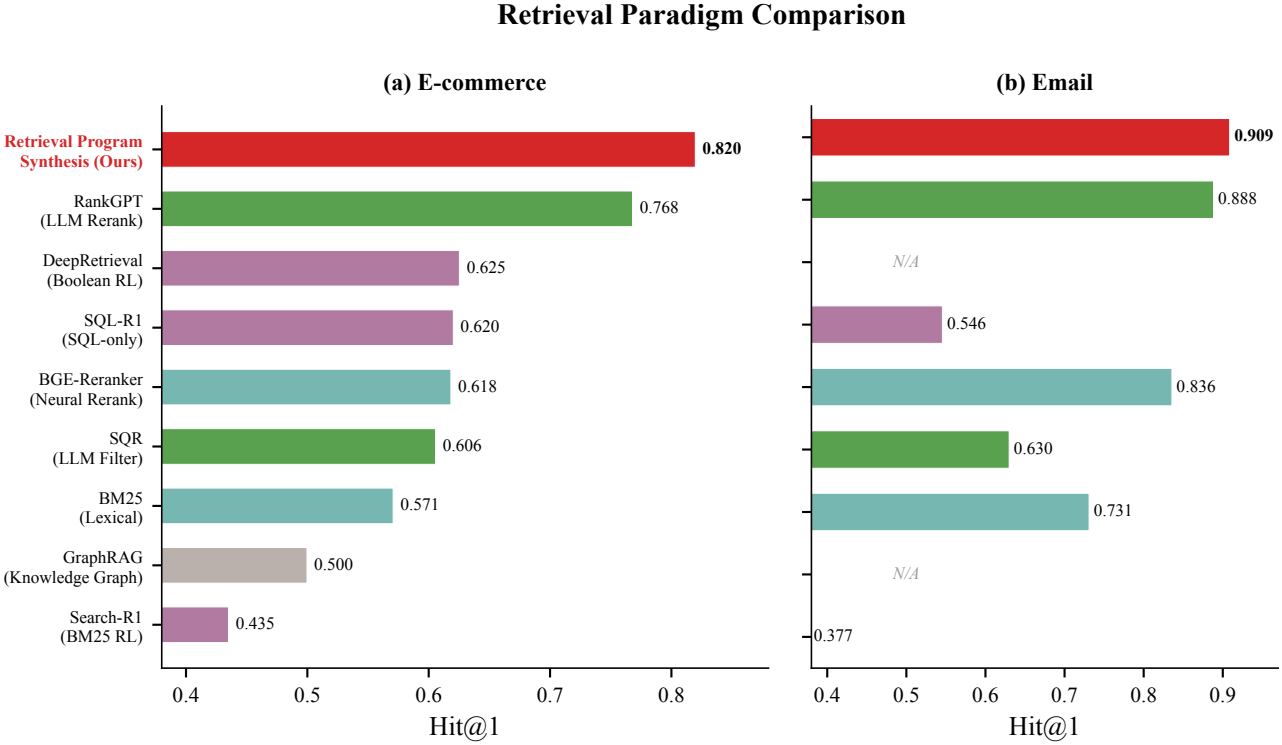

Retrieval Paradigm Comparison
(a) E-commerce
(b) Email
Retrieval Program Synthesis (Ours)
RankGPT (LLM Rerank)
DeepRetrieval (Boolean RL)
SQL-R1 (SQL-only)
BGE-Reranker (Neural Rerank)
SQR (LLM Filter)
BM25 (Lexical)
GraphRAG (Knowledge Graph)
Search-R1 (BM25 RL)
0.820
0.768
0.625
0.620
0.618
0.606
0.571
0.500
0.435
0.909
0.888
N/A
0.546
0.836
0.630
0.731
N/A
0.377
0.4
0.5
0.6
0.7
0.8
0.9
Hit@1
Hit@1